



CRACK THE VLSI INTERVIEW

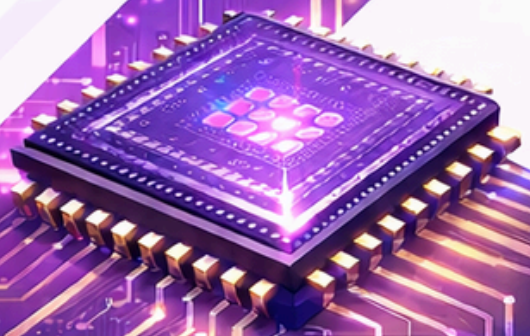
Advanced Aptitude, Logic & Design Screening Test

Think like an engineer. Answer like a pro.

INSIDE THIS PLAYBOOK

- ✓ Quantitative & Logical Challenges
- ✓ Clock, Timing & Pipeline Brain Teasers
- ✓ FSM Debugging & Design-centric Problems
- ✓ High-difficulty puzzles used in screenings

**Because your first round
decides your VLSI career.**



Question 1

A pipeline has 8 stages. Each stage takes 2.5 ns. New data enters every cycle. After how much time will the second output appear?

- A) 20 ns
- B) 17.5 ns
- C) 22.5 ns
- D) 25 ns

Question 2

A counter increments every clock. Clock = 200 MHz. How long to reach count 1000 starting from 0?

- A) 2 μ s
- B) 5 μ s
- C) 10 μ s
- D) 20 μ s

Question 3

A signal pattern is: 1 0 1 0 1 0 Clock is 100 MHz. What is signal frequency?

- A) 50 MHz
- B) 100 MHz
- C) 200 MHz
- D) 25 MHz

Question 4

A task is 60% parallelizable. If the parallel part is sped up by 4x, what is the overall speedup?

- A) 1.6x
- B) 1.9x
- C) 2.1x
- D) 2.5x

Question 5

A block runs at 500 MHz with 20% timing slack. What is the maximum achievable frequency?

- A) 550 MHz
- B) 600 MHz
- C) 625 MHz
- D) 700 MHz

Question 6

Total power is split: 40% clock, 30% logic, 30% memory. Clock power reduced by 50%. New total power is:

- A) 70%
- B) 75%
- C) 80%
- D) 85%

Question 7

Chip area reduces by 10%, yield improves from 70% to 85%. Which impacts cost more?

- A) Area reduction
- B) Yield improvement
- C) Both equal
- D) Cannot be determined

Question 8

If pipeline latency is 10 cycles and throughput is 1 item/cycle, how many items are in flight?

- A) 1
- B) 5
- C) 10
- D) Unlimited

Question 9

Output repeats every 6 cycles. Minimum FSM states?

- A) 2
- B) 3
- C) 6
- D) 12

Question 10

A signal toggles randomly with 50% probability per cycle. Average frequency is:

- A) 0
- B) $f/4$
- C) $f/2$
- D) f

Question 11

If a counter resets after reaching 15, how many bits are minimum?

- A) 3
- B) 4
- C) 5
- D) 15

Question 12

A pipeline produces one output every 3 cycles. Clock = 300 MHz. What is throughput?

- A) 100 M ops/s
- B) 300 M ops/s
- C) 900 M ops/s
- D) 3 ops/s

Question 13

A clock divider generates output HIGH for 2 cycles and LOW for 1 cycle repeatedly. What is duty cycle?

- A) 33.3%
- B) 50%
- C) 66.7%
- D) 75%

Question 14

A mod-7 counter implemented using 3 flip-flops enters state 111. What should ideally happen?

- A) Continue counting
- B) Lock in that state
- C) Reset to 000
- D) Go to 001

Question 15

Two pipelines A and B have same throughput. Pipeline A has 5 stages, B has 10 stages. Which is true?

- A) A always faster
- B) B always faster
- C) Same performance always
- D) Depends on workload

Question 16

A FIFO has depth 8. Write happens every cycle. Read happens once every 2 cycles. After how many cycles will FIFO overflow?

- A) 8
- B) 12
- C) 14
- D) 16

Question 17

A pipeline has variable stage delays: [1 ns, 1 ns, 4 ns, 1 ns]. The clock period must be at least:

- A) 1 ns
- B) 2 ns
- C) 4 ns
- D) 7 ns

Question 18

An FSM has 5 states but uses 3 flip-flops. How many unused states exist?

- A) 1
- B) 2
- C) 3
- D) 5

Question 19 (Think Carefully)

A signal pattern repeats every 9 cycles. FSM outputs depend on both state and input. Minimum number of states required?

- A) 4
- B) 5
- C) 9
- D) Cannot be determined

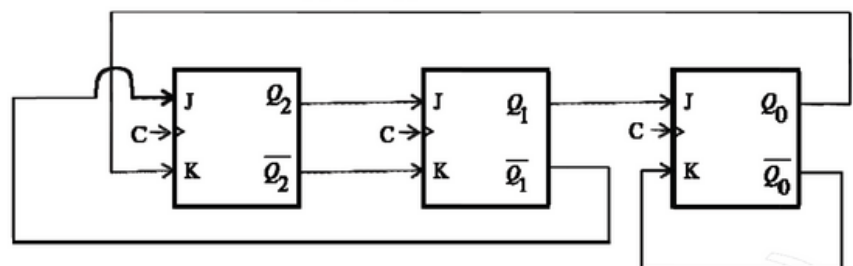
Question 20

A clock is divided by an odd number. Which property is hardest to guarantee?

- A) Frequency accuracy
- B) Duty cycle
- C) Phase noise
- D) Power

Question 21

The above sequential circuit is built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycle is



- A) 001, 010, 011
- B) 111, 110, 101
- C) 100, 110, 111
- D) 100, 011, 001



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